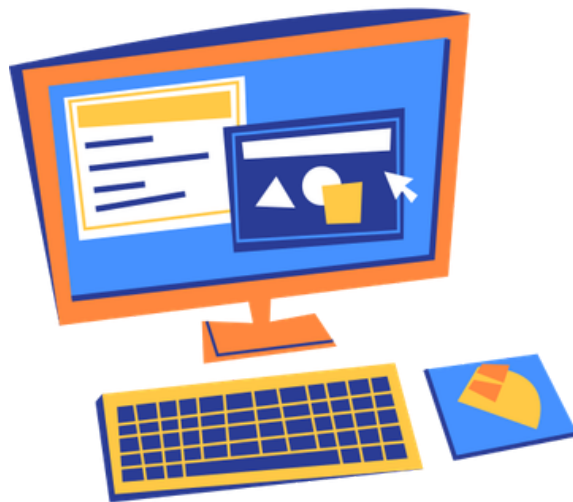


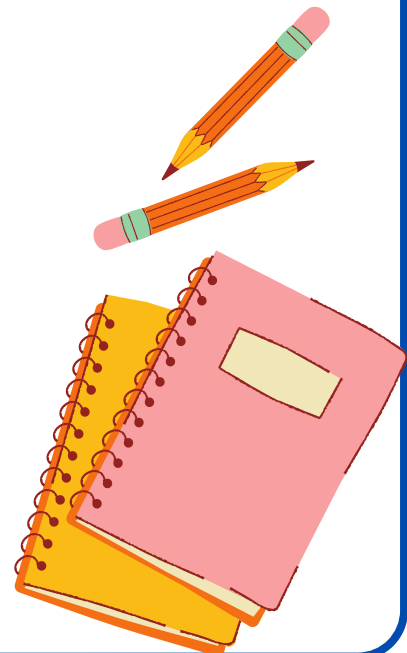


CS Workbook II

Chapter 3 OOP



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Object Oriented Programming

- ★ Firstly programming was used which was used to describe “what” needs to be process for the program, This was **declarative programming** They require functions and expressions but no states

Example

(sql, html, some js code,)

1) some database codes to get or put into tables.

- ★ Now a days we use complex systems ,**imperative programming languages** ,which require processing on basis of context “how” ,”how” the computer program will work

- ★ Procedural programming language has functions, pointers, eg C embedded systems like microwave oven

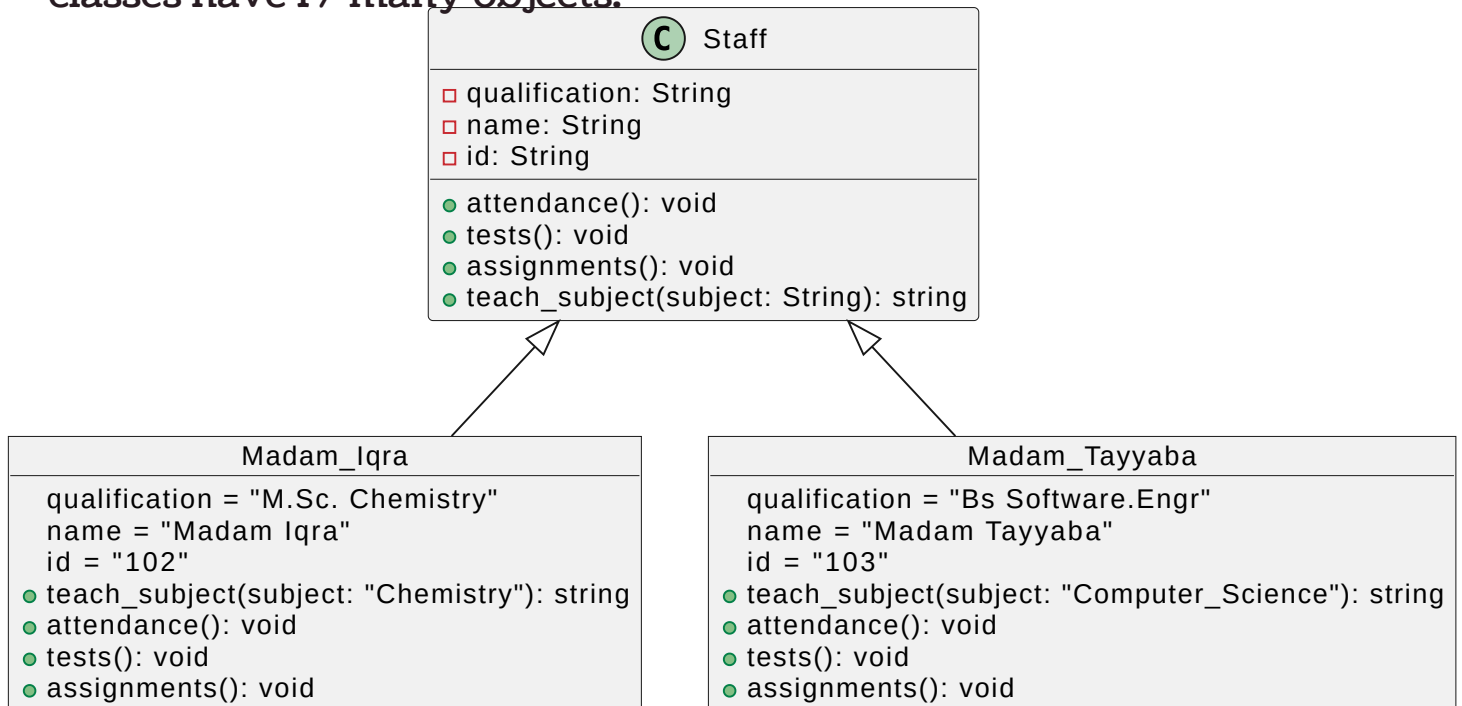
- ★ where there is some procedures ,some states, some objects .**Oop** is a type of imperative language programming model

Example

(C++, Python)

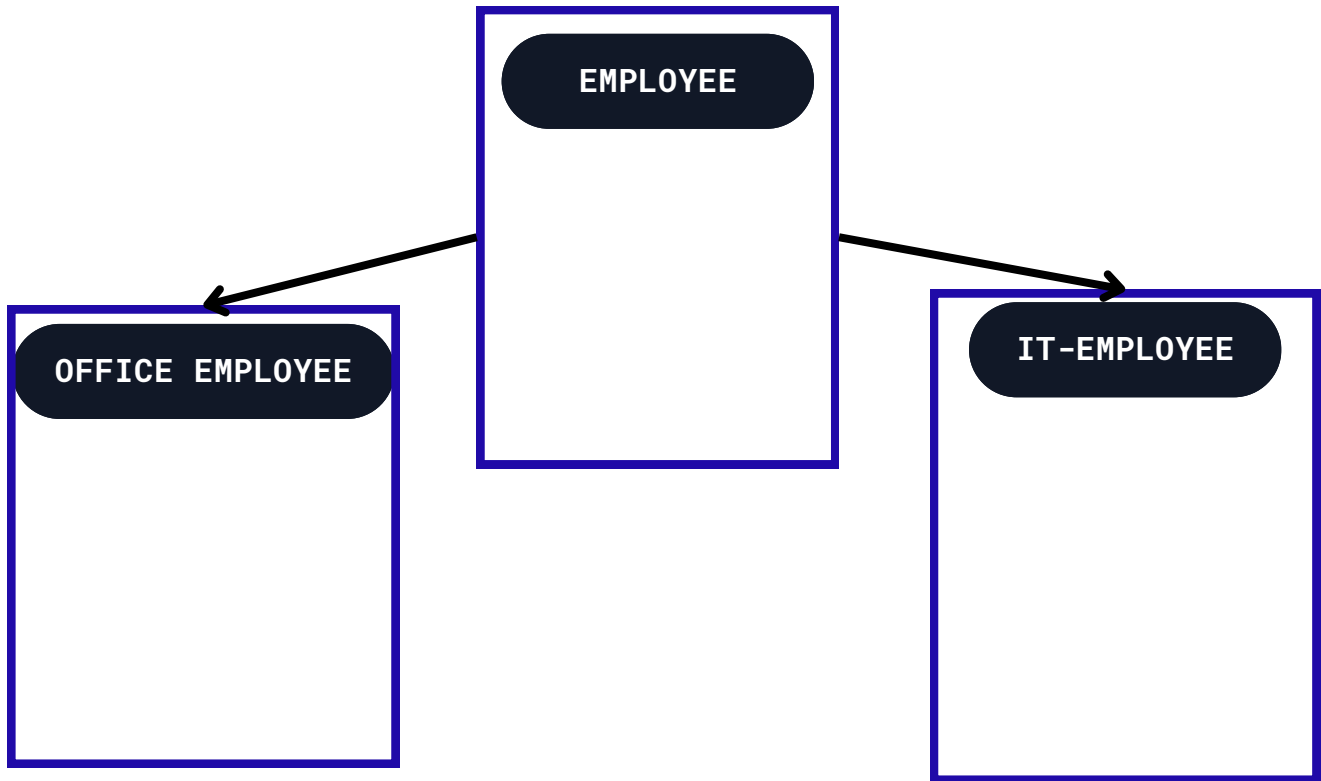
1) using variables and classes for writing a program

2) each class(noun) has members(adjectives) and functions(verbs)
classes have 1 / many objects.

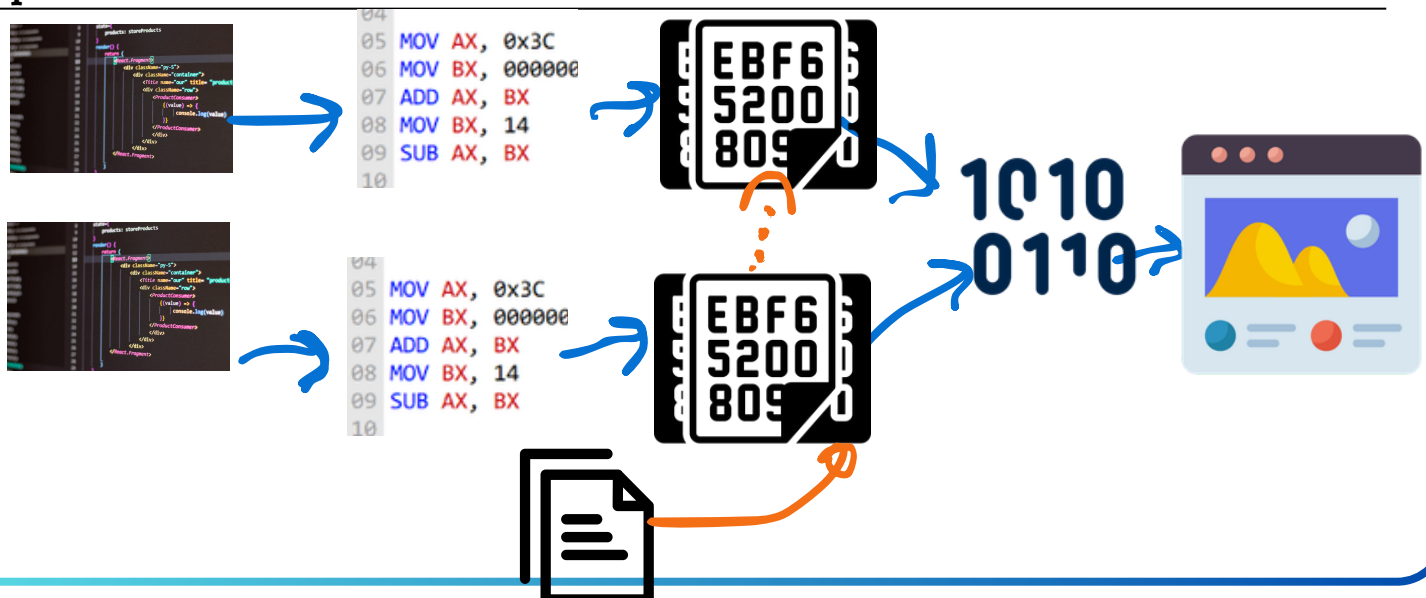


Brain Storming Exercise

Question: Understand Oop in real life scenario eg for office employees ?
employee should have members as-well as functions employee may have
members like employee_id, employee_name and functions like
computer_work(), enter_attendance() ?



- 1 C++ (source codes(header files preprocess)--assembly codes--object codes(hexadecimal memory addresses 0-9 A-F) ----executable code--process)



Program
set of instructions that perform a specific task when executed by the computer



C++ programs are source code, assembly code, object code and executable code.

when i click compile on c++ compiler "compile" it creates programs

Process
program in its running state is a process



C++ process are executable code which are running on hardware

when i click "run" on c++ compiler it creates processes

1 Source code is C++ program is also called

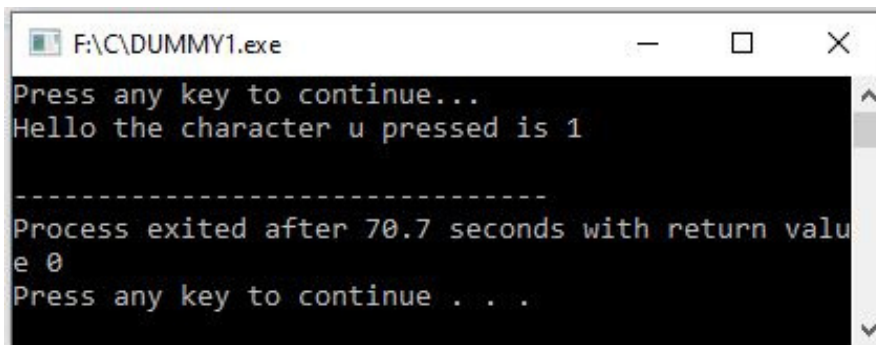
language and object code is called language

Lets move towards compiler What would be the output for program?

```
DUMMY1.cpp x
1  #include <iostream>
2  #include <conio.h>
3  using namespace std;
4  int main() {
5      char ch;
6      int cin;
7      cout << "Press any key to continue...\n";
8
9      ch=getch();
10
11      cout << "Hello the character u pressed is " << ch << std::endl;
12
13      return 0;
14  }
15
```

Preprocessor directive :line 1 and 2

Executable statements/Main program



```
F:\C\DUMMY1.exe
Press any key to continue...
Hello the character u pressed is 1

-----
Process exited after 70.7 seconds with return value 0
Press any key to continue . . .
```



What would happen if I use `cin >>ch;` instead of line 9?

